



Knowledge Distillation on Graphs: A survey

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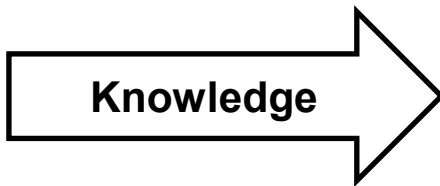
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Knowledge Distillation

- Technique to transfer knowledge from a teacher model to a student model

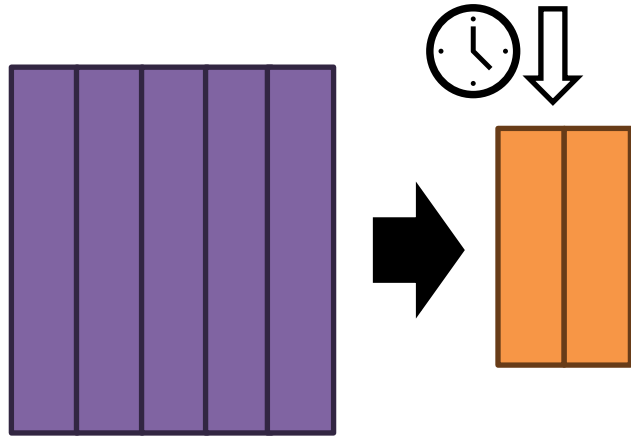


Teacher

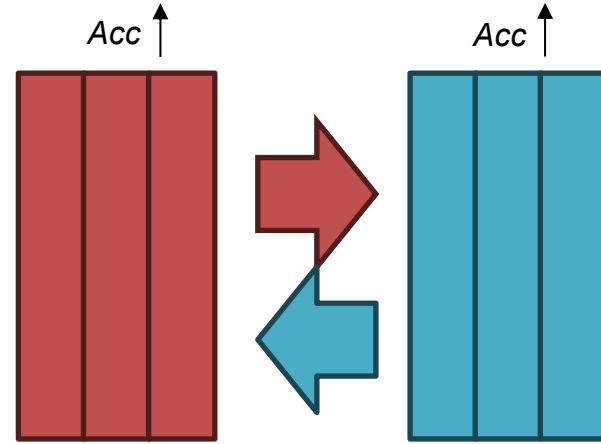


Student

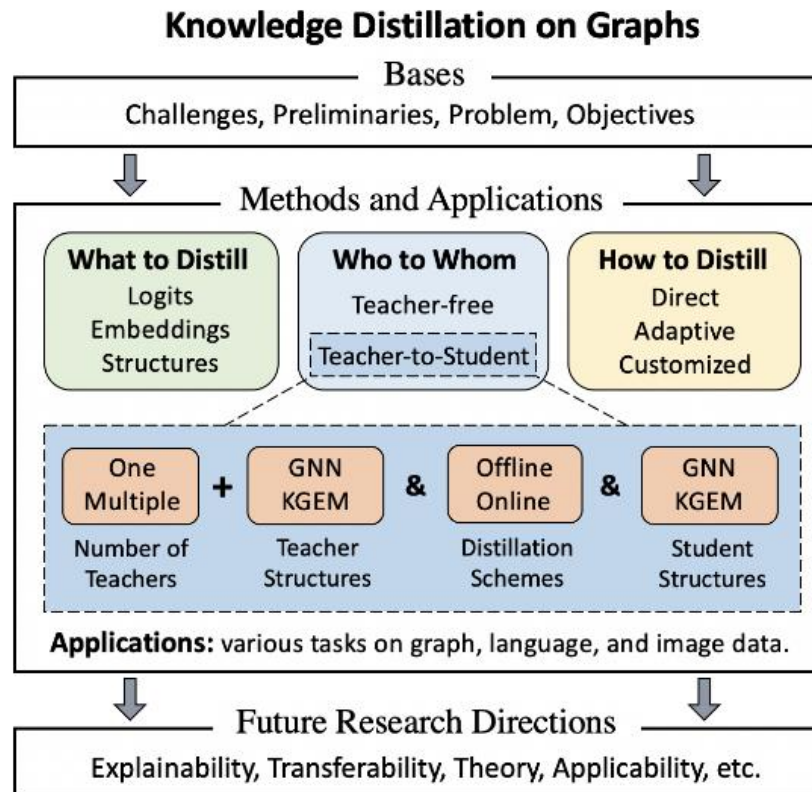
Objectives



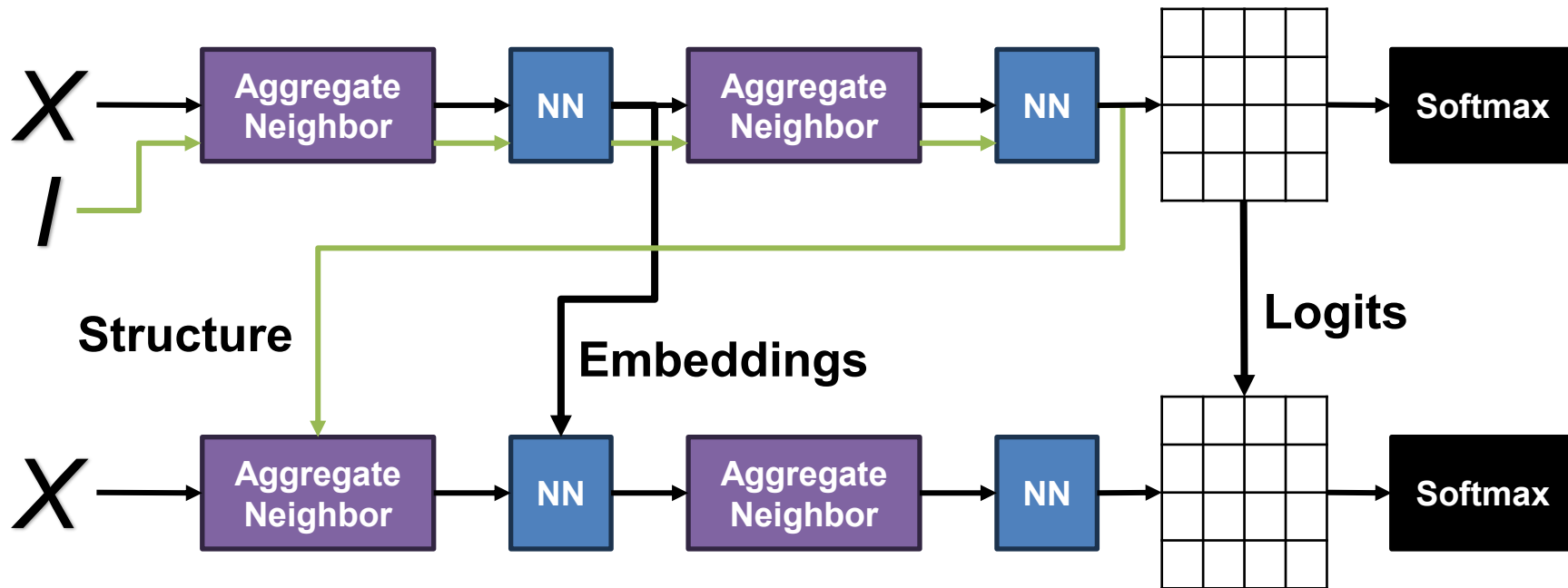
Model Compression



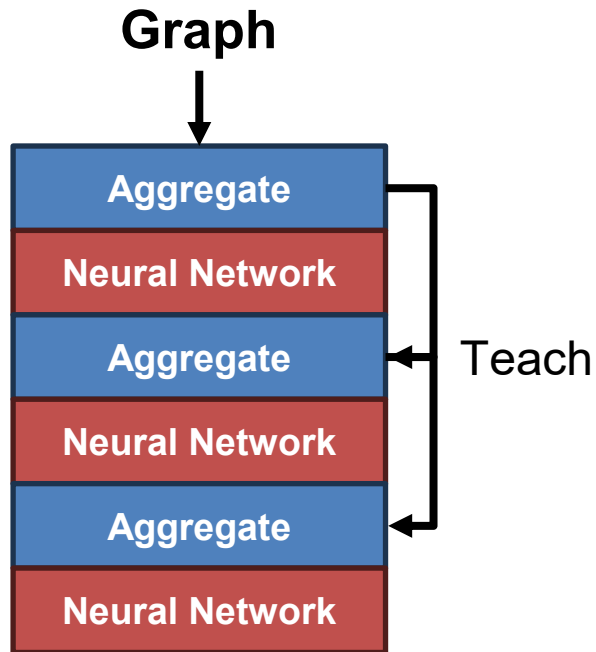
Performance Improvement



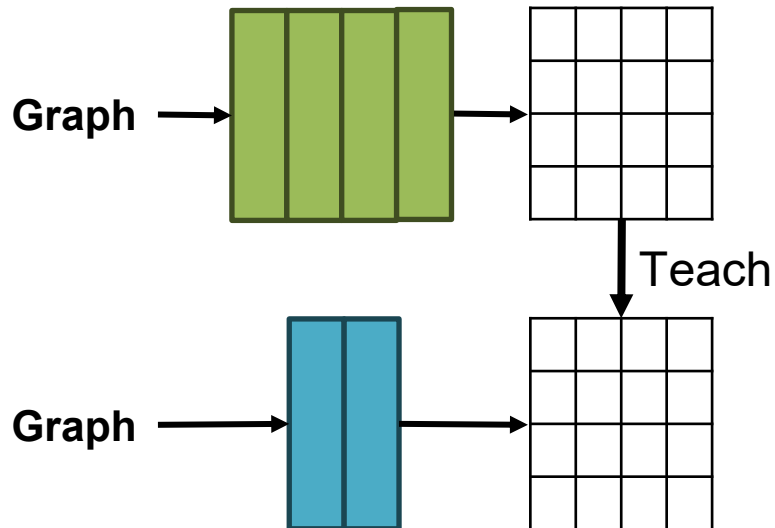
What to Distill



Who to Whom

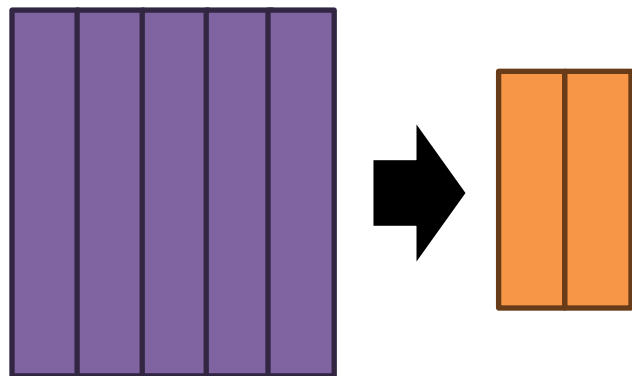


Teacher-Free

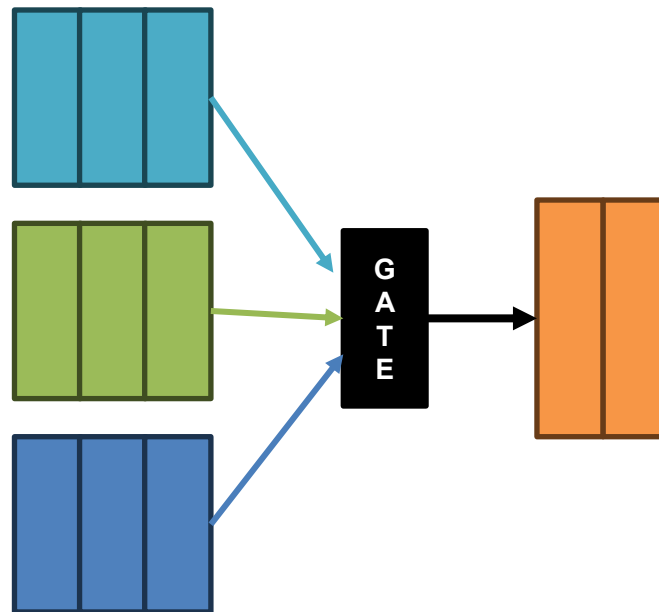


Teacher-to-Student

Teacher-to-Student – Number of Teachers

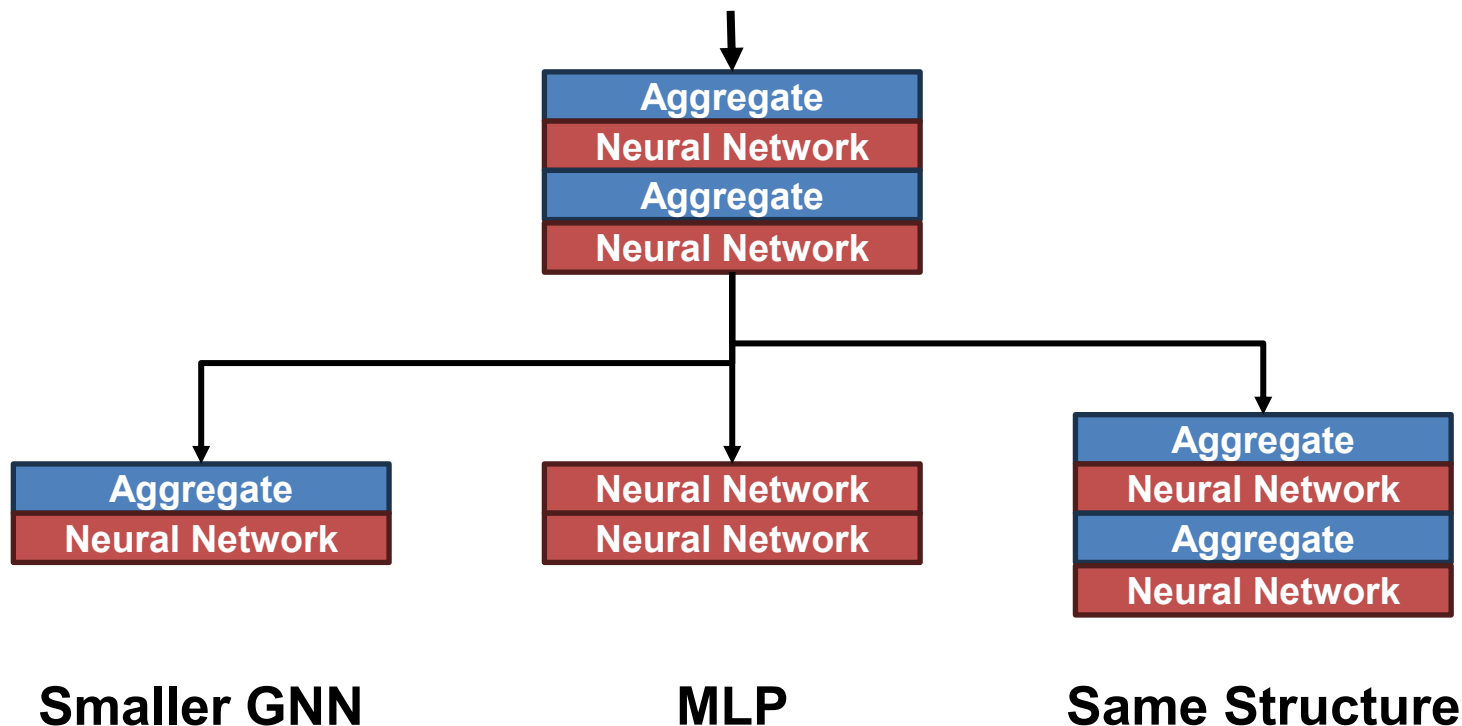


One Teacher

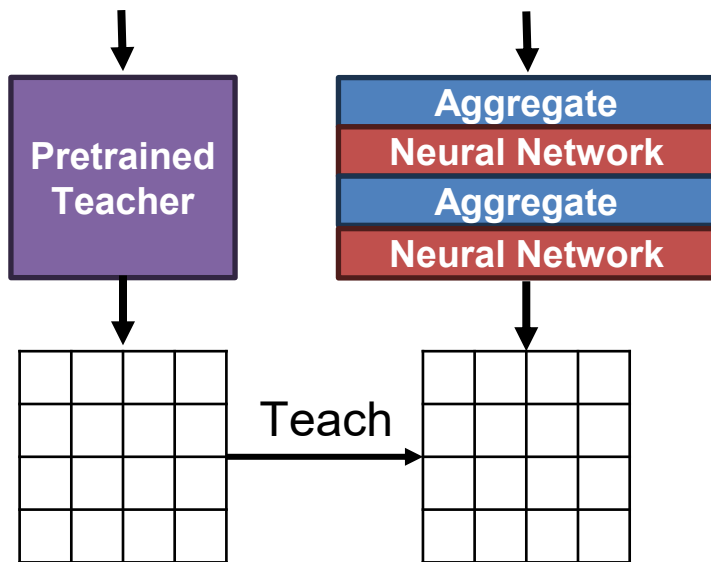


Multiple Teachers

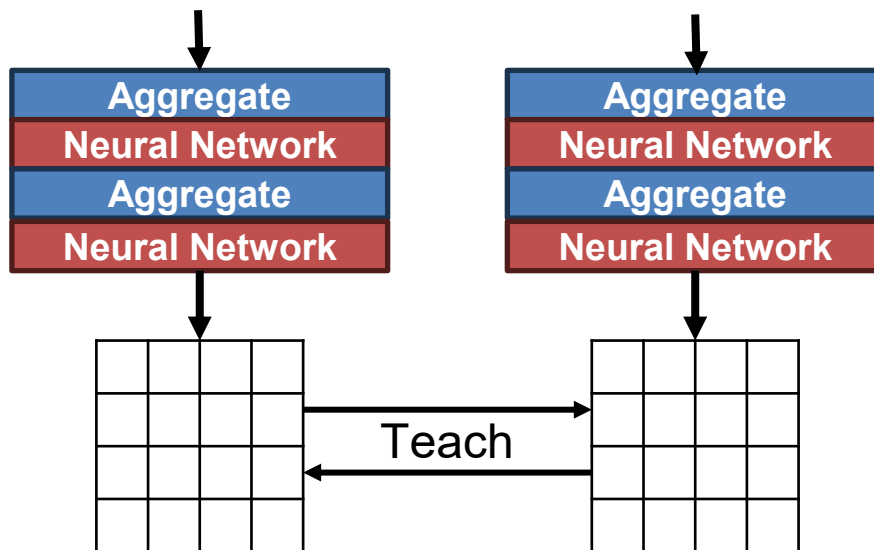
Teacher-to-Student – Teacher and Student Structures



Teacher-to-Student – Offline and Online

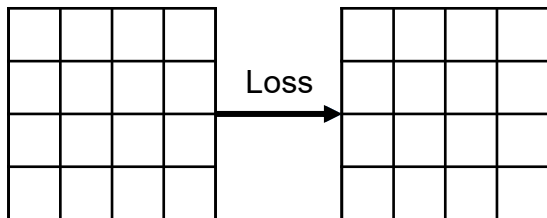


Offline Distillation

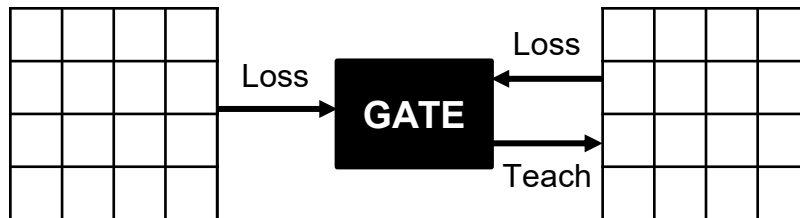


Online Distillation

How to Distill



Direct Distillation



Adaptive Distillation



Customized Distillation



LightHGNN: Distilling Hypergraph Neural Networks Into MLPs For 100x Faster Inference

Yifen Feng¹, Yihe Luo¹, Shihui Ying², Yue Gao¹

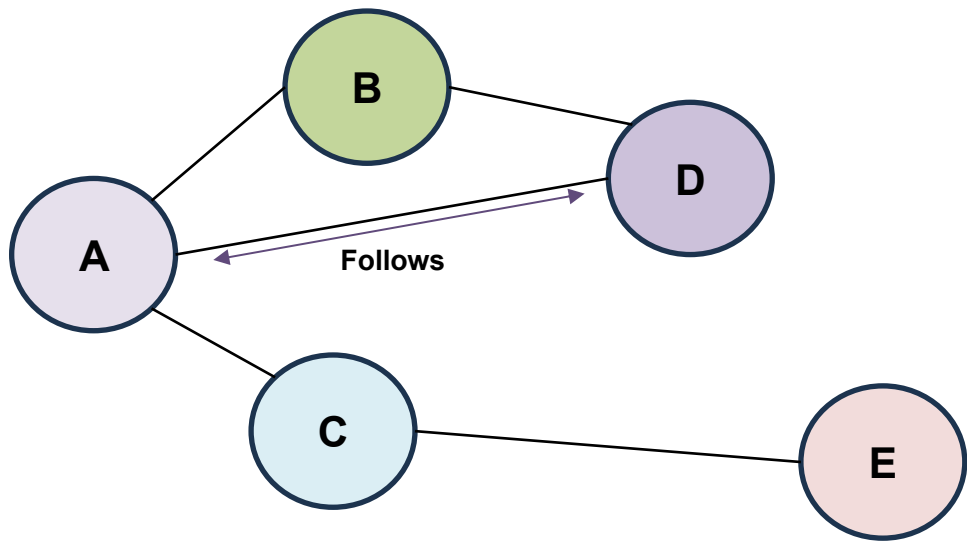
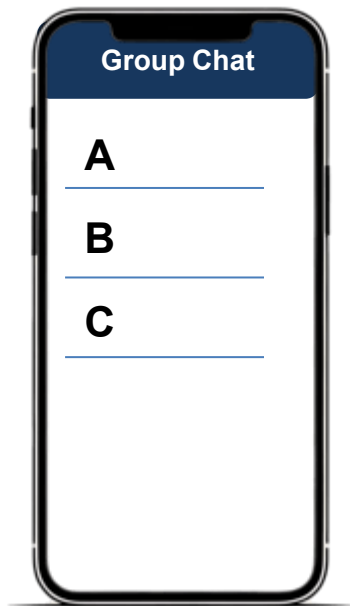
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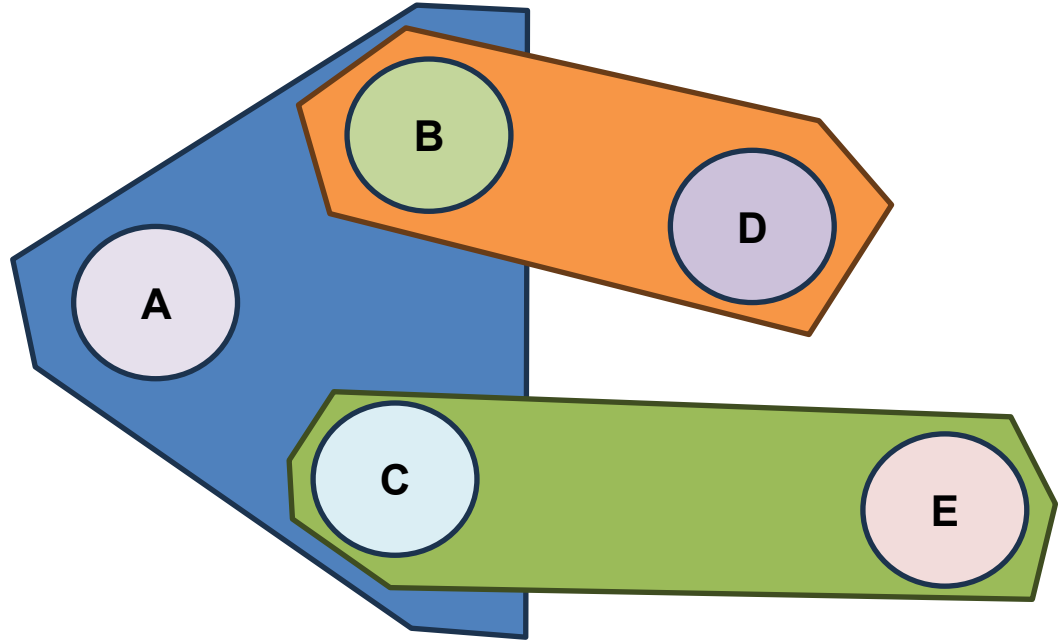
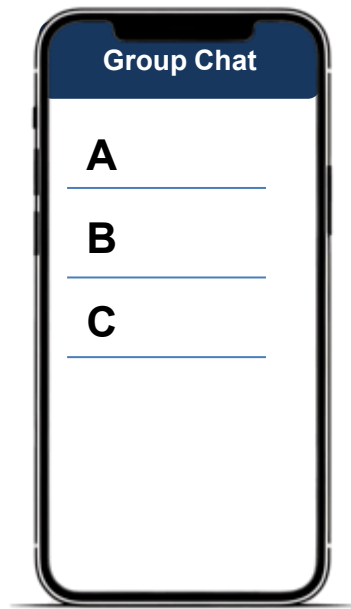
Limitation of GNNs

- GNNs struggle to capture group-level, higher-order relationships



What is a Hypergraph?

- Hypergraphs represent multi-node interactions that connect sets of nodes



HGNN vs GCN

- HGNN simply changes hyperedges to edges and make it to GCN

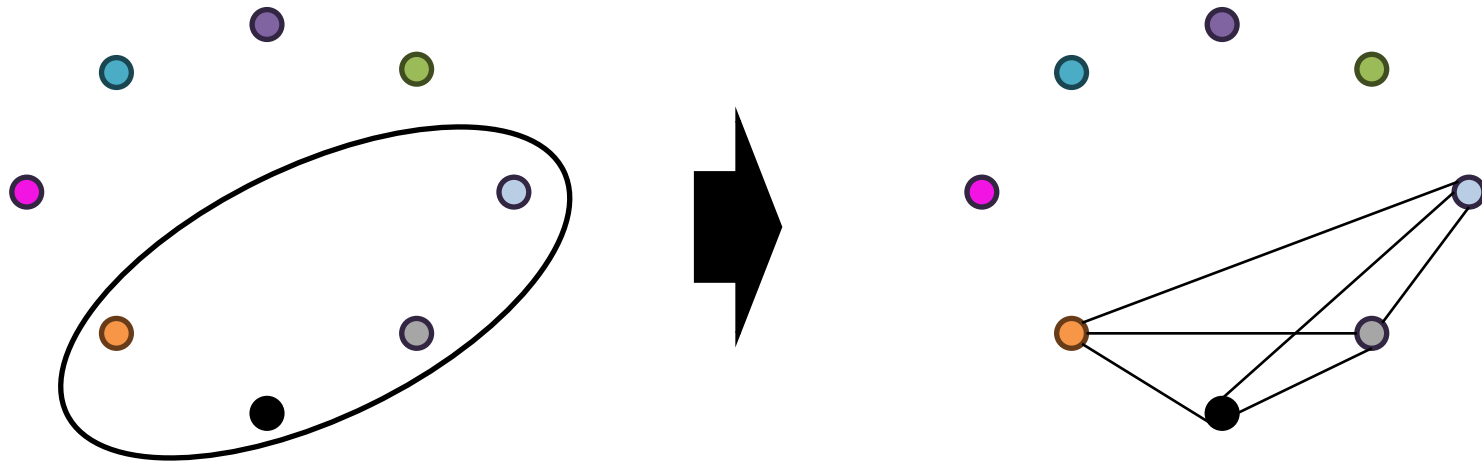
HGNN

$$Y = D_v^{-1/2} \underbrace{H W D_e^{-1} H^T}_{\text{Equal Role}} D_v^{-1/2} X \Theta$$

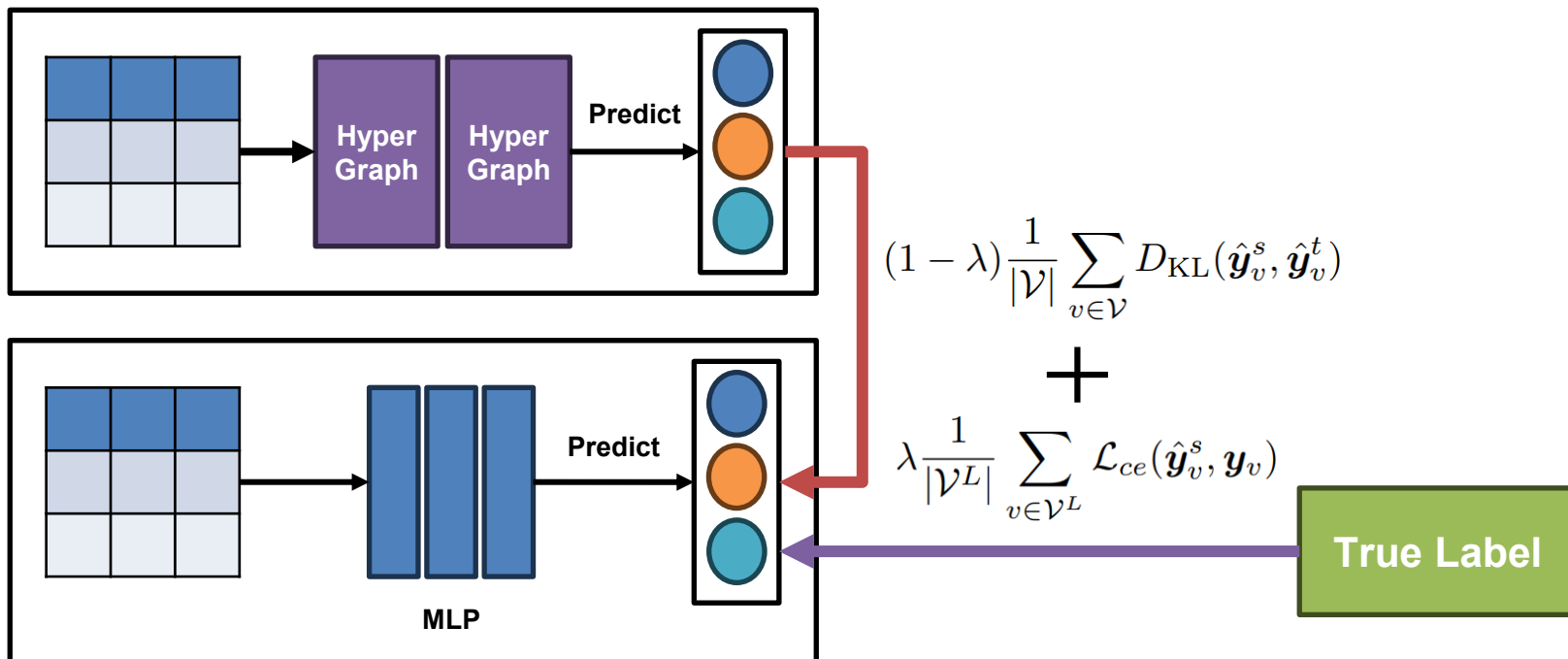
GCN

$$Z = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} X \Theta$$

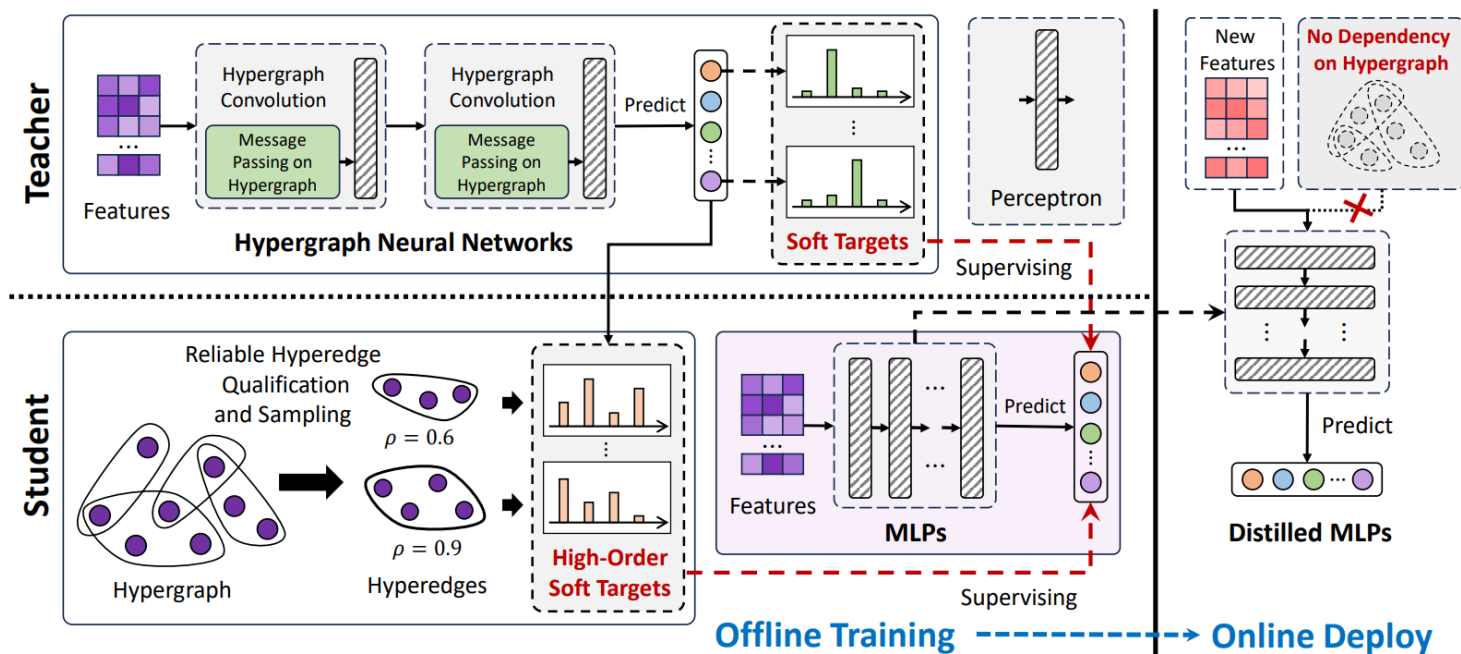
HGNN vs GCN (Cont.)



- Directly distills hypergraph prediction soft labels to student MLP



- Additionally inject the reliable high-order information into the MLPs



- Add noise and compare information entropy to quantify the relevance of the hyperedge

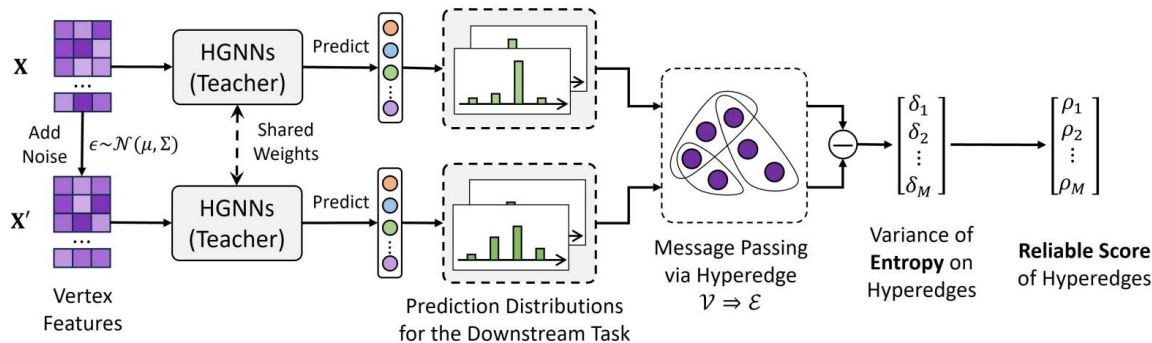


Figure 3: The illustration of reliable hyperedge qualification.

$$\mathcal{H}(p) = -\sum_i p_i \log(p_i)$$

$$\epsilon \sim \mathcal{N}(\mu, \Sigma)$$

$$\delta_e = \mathbb{E}_{\epsilon \sim \mathcal{N}(\mu, \Sigma)} \left\| \frac{1}{|e|} \sum_{v \in e} \mathcal{H}(\hat{y}'_v) - \frac{1}{|e|} \sum_{v \in e} \mathcal{H}(\hat{y}_v) \right\|_2,$$

$$\text{where } Y' = f_\theta(\epsilon X, H) \quad \text{and} \quad Y = f_\theta(X, H)$$

- The bigger the reliable score ρ_e of the hyperedge, the more robust the knowledge, and therefore increase the possibility to be left

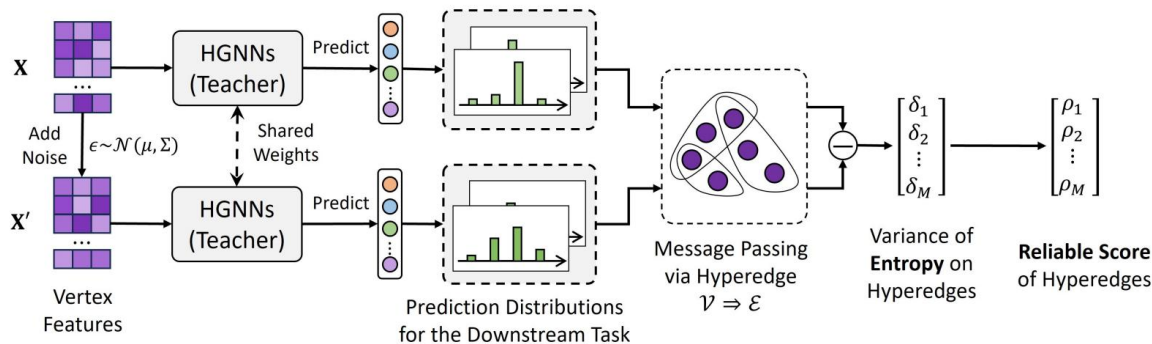


Figure 3: The illustration of reliable hyperedge qualification.

$$\rho_e = 1 - \frac{\delta_e}{\delta_{\max}} \quad p(s_i | \rho_{e_i}) \sim \text{Bernoulli}(\rho_{e_i})$$

- Add high-order information KD divergence to the loss function

$$\mathbf{z}_e^s = \frac{1}{|e|} \sum_{v \in \mathcal{N}_v(e)} \hat{\mathbf{y}}_v^s \quad \text{and} \quad \mathbf{y}_e^t = \frac{1}{|e|} \sum_{v \in \mathcal{N}_v(e)} \hat{\mathbf{y}}_v^t \quad \text{for } e \in \mathcal{E}',$$

Average the soft labels of connected vertices of the hyperedge e

$$\mathcal{L}_{hc} = \frac{1}{|\mathcal{E}'|} \sum_{\substack{e \in \mathcal{E}' \\ e_i \sim p(s_i | \rho_{e_i})}} D_{\text{KL}}(\alpha(\mathbf{z}_e^s / \tau), \alpha(\mathbf{y}_e^t / \tau)),$$

Add KL Divergence of the predicted labels and high-order labels

$$\mathcal{L}_{DH^+} = \lambda \frac{1}{|\mathcal{V}^L|} \sum_{v \in \mathcal{V}^L} \mathcal{L}_{ce}(\hat{\mathbf{y}}_v^s, \mathbf{y}_v) + (1 - \lambda) \left(\frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} D_{\text{KL}}(\hat{\mathbf{y}}_v^s, \hat{\mathbf{y}}_v^t) + \mathcal{L}_{hc} \right)$$

Add it to the final loss function

Experiments – Performance

Dataset	Setting	MLPs	HGNN	LightHGNN	LightHGNN ⁺	Δ_{MLP}	Δ_{HGNN}
<i>News20</i>	Prod.	63.86 $\pm$ 3.01	75.18 $\pm$ 1.65	75.81 $\pm$ 1.28	74.56 $\pm$ 1.34	11.95	+0.63
	Tran.	63.80 $\pm$ 3.03	75.08 $\pm$ 1.44	75.81 $\pm$ 1.23	74.58 $\pm$ 1.24	12.01	+0.73
	Ind.	64.10 $\pm$ 2.97	75.13 $\pm$ 2.01	75.82 $\pm$ 1.61	75.44 $\pm$ 1.94	11.71	+0.68
<i>CA-Cora</i>	Prod.	50.73 $\pm$ 1.43	71.01 $\pm$ 3.19	70.33 $\pm$ 3.49	71.46 $\pm$ 2.13	20.72	+0.45
	Tran.	50.75 $\pm$ 1.64	70.80 $\pm$ 3.25	71.62 $\pm$ 4.29	72.49 $\pm$ 2.13	21.73	+1.68
	Ind.	50.67 $\pm$ 1.44	70.83 $\pm$ 2.83	65.14 $\pm$ 2.95	67.34 $\pm$ 3.36	16.67	-3.48
<i>CC-Cora</i>	Prod.	50.73 $\pm$ 1.43	68.20 $\pm$ 3.89	68.29 $\pm$ 4.47	67.89 $\pm$ 3.58	17.55	+0.09
	Tran.	50.75 $\pm$ 1.64	68.26 $\pm$ 3.92	69.00 $\pm$ 4.16	68.70 $\pm$ 3.32	18.24	+0.73
	Ind.	50.67 $\pm$ 1.44	66.00 $\pm$ 4.55	65.46 $\pm$ 5.87	64.61 $\pm$ 4.69	14.79	-0.53
<i>CC-Citeseer</i>	Prod.	54.41 $\pm$ 1.36	64.02 $\pm$ 0.92	62.90 $\pm$ 1.95	64.11 $\pm$ 0.85	9.69	+0.09
	Tran.	54.42 $\pm$ 1.52	63.74 $\pm$ 0.75	63.30 $\pm$ 1.92	64.53 $\pm$ 0.63	10.10	+0.79
	Ind.	54.36 $\pm$ 1.14	63.51 $\pm$ 1.34	61.31 $\pm$ 2.21	61.93 $\pm$ 2.01	7.56	-1.58
<i>DBLP-Paper</i>	Prod.	63.23 $\pm$ 1.48	71.52 $\pm$ 1.31	71.14 $\pm$ 1.23	71.69 $\pm$ 1.44	8.46	+0.16
	Tran.	62.97 $\pm$ 1.69	70.75 $\pm$ 1.49	70.88 $\pm$ 1.29	71.40 $\pm$ 1.50	8.42	+0.65
	Ind.	64.25 $\pm$ 1.75	72.72 $\pm$ 2.32	72.22 $\pm$ 2.08	72.86 $\pm$ 2.33	8.61	+0.14
<i>DBLP-Term</i>	Prod.	63.56 $\pm$ 1.15	81.08 $\pm$ 2.51	78.39 $\pm$ 3.22	78.32 $\pm$ 2.70	14.83	-2.69
	Tran.	63.37 $\pm$ 1.17	81.23 $\pm$ 2.39	78.54 $\pm$ 3.08	78.58 $\pm$ 2.73	15.21	-2.64
	Ind.	64.30 $\pm$ 1.50	81.56 $\pm$ 2.75	77.79 $\pm$ 4.15	77.28 $\pm$ 3.29	13.48	-3.77
<i>DBLP-Conf</i>	Prod.	63.56 $\pm$ 1.15	94.15 $\pm$ 0.19	89.48 $\pm$ 0.52	89.50 $\pm$ 0.49	25.94	-4.64
	Tran.	63.37 $\pm$ 1.17	94.08 $\pm$ 0.32	91.12 $\pm$ 0.76	91.20 $\pm$ 0.74	27.83	-2.87
	Ind.	64.30 $\pm$ 1.50	94.21 $\pm$ 0.52	82.93 $\pm$ 1.15	82.68 $\pm$ 0.57	18.62	-11.27
<i>IMDB-AW</i>	Prod.	41.05 $\pm$ 2.49	50.29 $\pm$ 1.58	49.10 $\pm$ 1.64	49.12 $\pm$ 2.02	8.05	-1.17
	Tran.	41.16 $\pm$ 2.67	49.46 $\pm$ 1.43	49.39 $\pm$ 1.63	49.68 $\pm$ 1.86	8.51	+0.21
	Ind.	40.61 $\pm$ 1.95	52.05 $\pm$ 2.68	47.96 $\pm$ 2.12	47.58 $\pm$ 3.21	7.35	-4.08
Avg. Rank/Avg.		4.0	1.5	2.3	1.9	14.86	-1.45

- Higher score indicates a lower relevance to the topology

Model	Graph Datasets			Hypergraph Datasets			Avg.
	Cora	Pubmed	Citeseer	CA-Cora	DBLP-Paper	IMDB-AW	
MLPs	3.78	2.06	2.64	2.76	1.02	1.18	2.24
GCN	0.25	0.32	0.09	0.08	0.12	0.08	0.15
HGNN	0.31	0.27	0.13	0.10	0.12	0.04	0.16
LightHGNN	1.20	1.50	0.33	0.94	0.69	0.71	0.89
LightHGNN ⁺	1.58	1.95	0.64	1.15	0.73	0.77	1.14

$$\mathcal{S} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{\sum_{v \in e} \|\mathbf{x}_v - \mathbf{x}_e\|_2}{d_e(e)} \quad \text{and} \quad \mathbf{x}_e = \frac{1}{d_e(e)} \sum_{v \in e} \mathbf{x}_v$$



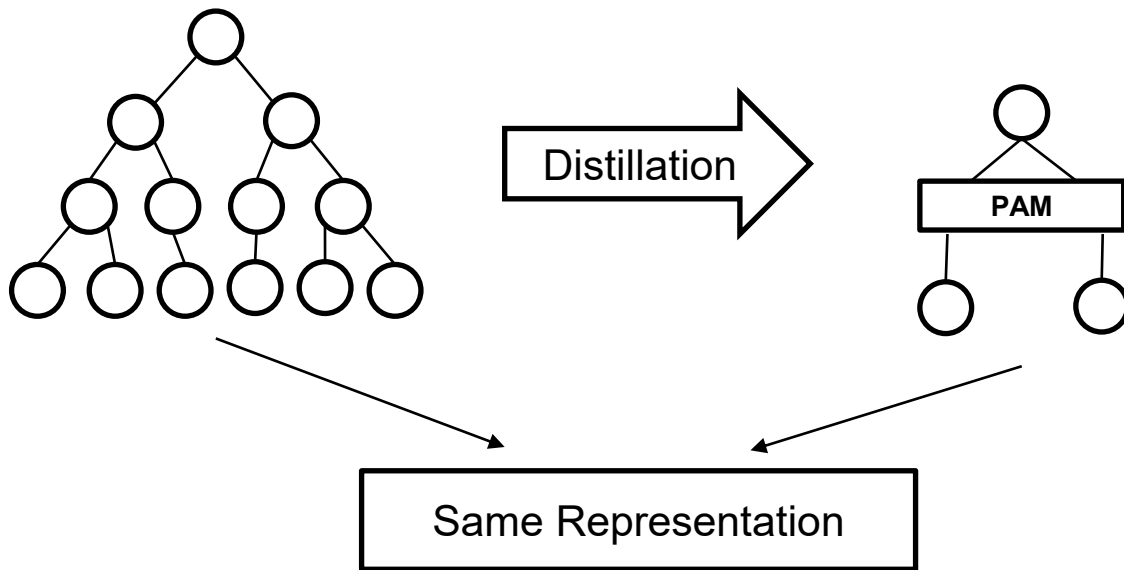
TinyGNN: Learning Efficient Graph Neural Networks

Bencheng Yan, Chaokun Wang, Gaoyang Guo, Yunkai Lou
Tsinghua University

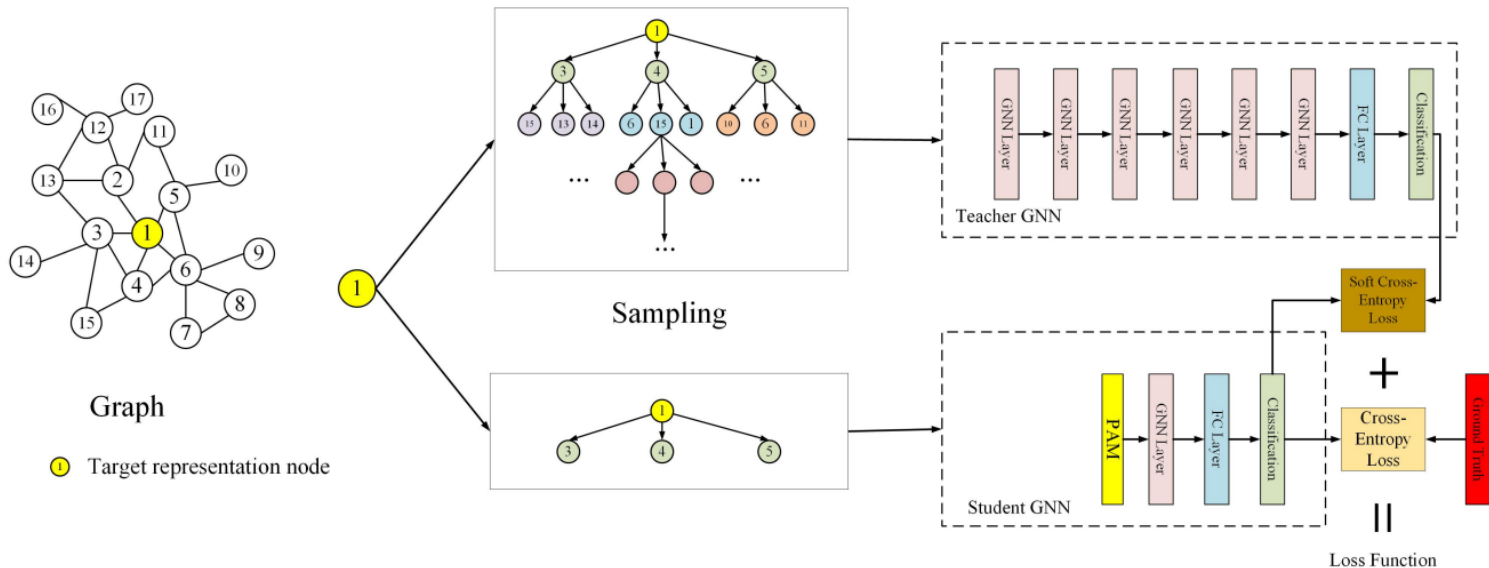
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Goal of TinyGNN

- Make shallow GNNs to have similar representation powers as deep GNNs

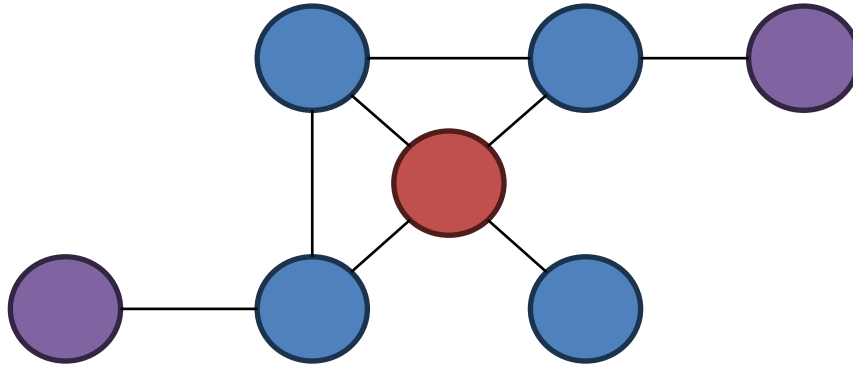


Full Structure of TinyGNN



Relations Between Neighbor Nodes

- Neighborhood nodes are connected to each other in maximum 2 hops



Relations Between Neighbor Nodes (cont.)

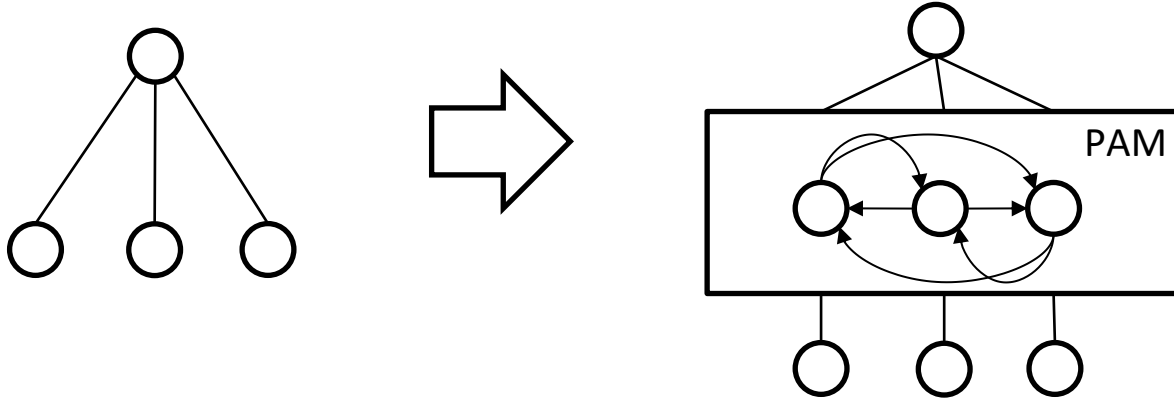
- Many peer nodes are also directly connected to each other

Table 1: Statistics of direct link rate among peer nodes.

Data Sets	Facebook	Chameleon	Squirrel	AliGraph
rate (%)	32.81	42.94	46.91	20.69

Peer-Aware Module for Student Model

- Add module to update representation of nodes considering its peers



- Apply self-attention to aggregate peer information

$$\alpha_{i,j} = \frac{\exp(a_{i,j})}{\sum_k \exp(a_{i,k})}$$
$$a_{i,j} = \frac{\text{dot}(W_a h_j, W_a h_i)}{\sqrt{d}}$$

$$h_i^* = \sum_j \alpha_{i,j} \cdot (W_v \cdot h_j)$$

Full Algorithm of the Model

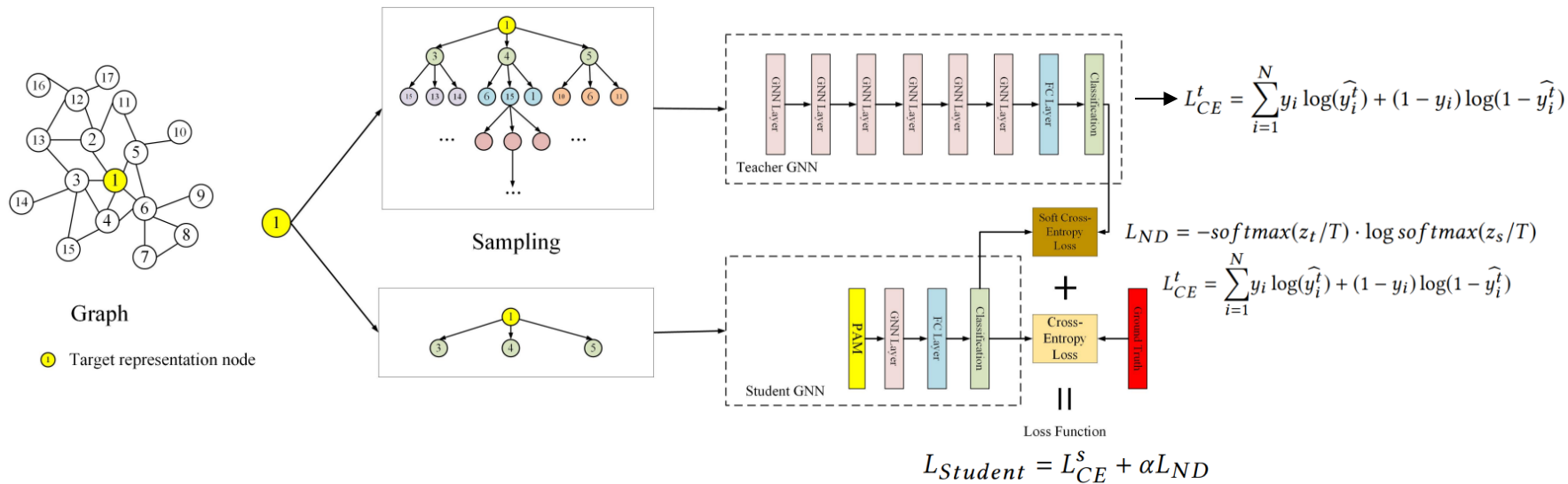


Table 4: The inference time for each model on different data sets.

model	Facebook	Chameleon	Squirrel	AliGraph
GNN_3 (Teacher)	18.1999 (1 $\times$)	1.2715 (1 $\times$)	3.2022 (1 $\times$)	106.7618 (1 $\times$)
GNN_2 (Base)	0.8654 (21.03 $\times$)	0.0856 (14.85 $\times$)	0.164 (19.53 $\times$)	3.4994 (30.51 $\times$)
GNN_1 (Base)	0.2703 (67.33 $\times$)	0.03 (40.50 $\times$)	0.0447 (67.13 $\times$)	0.5081 (210.12 $\times$)
$TinyGNN_1$	0.4314 (42.19 $\times$)	0.0332 (38.30 $\times$)	0.0897 (35.70 $\times$)	0.8434 (126.59 $\times$)
$TinyGNN_2$	2.3542 (7.73 $\times$)	0.1059 (12.00 $\times$)	0.3112 (10.29 $\times$)	5.0072 (21.32 $\times$)

Experiments – Different Teacher and Student Models

Table 5: Performance comparison with different teacher and different student GNNs. The best results for GNNs with the same layer number are in bold.

		Facebook		Chameleon		Squirrel		AliGraph	
Base		Micro-f1(%)	Macro-f1(%)	Micro-f1(%)	Macro-f1(%)	Micro-f1(%)	Macro-f1(%)	Micro-f1(%)	Macro-f1(%)
GNN_1		82.75	81.65	36.88	35.79	29.83	29.24	33.77	25.22
GNN_2		87.69	86.83	38.73	36.92	30.47	30.08	49.98	43.45
GNN_3		89.43	88.69	39.12	38.16	30.68	30.41	60.34	53.75
GNN_4		90.21	89.53	38.73	36.57	29.70	28.94	64.37	58.35
Teacher	Student								
GNN_4	$TinyGNN_1$	84.99	83.75	39.22	37.72	32.86	32.48	42.67	32.39
GNN_3	$TinyGNN_1$	84.56	83.41	39.71	38.46	32.09	32.00	42.34	32.37
GNN_2	$TinyGNN_1$	83.04	81.73	38.24	36.12	32.22	31.66	41.43	31.85
GNN_4	$TinyGNN_2$	89.89	89.24	40.68	39.22	32.44	32.29	61.28	54.26
GNN_3	$TinyGNN_2$	89.40	88.66	41.17	40.01	33.33	33.17	61.12	54.17
GNN_4	$TinyGNN_3$	90.83	90.25	41.46	40.46	31.58	31.40	67.07	60.83



DistillHGNN: A Knowledge Distillation Approach For High-Speed Hypergraph Neural Networks

Saman Forouzandeh, Parham Moradi & Mahdi Jalili

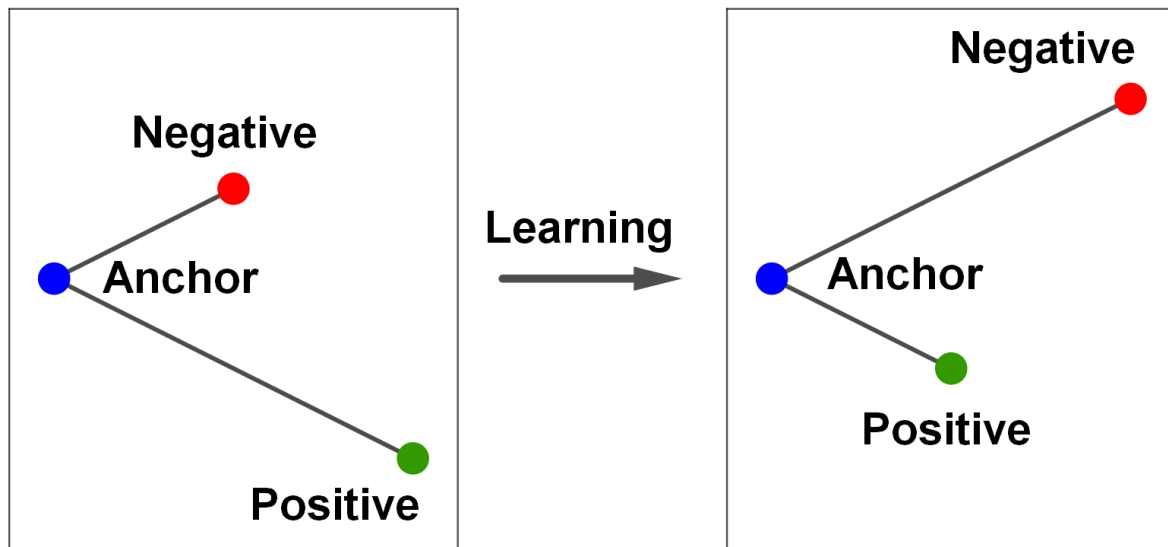
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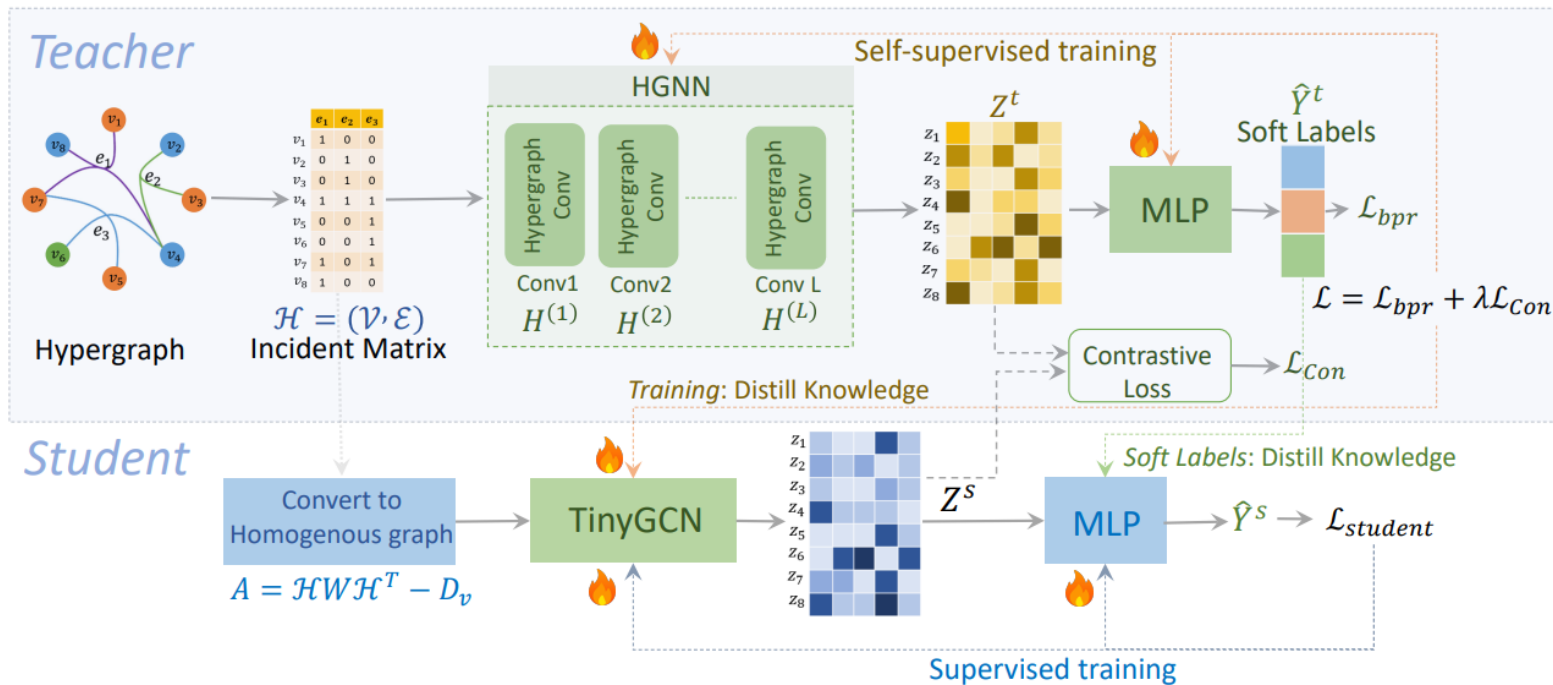
Joohee Cho

Contrastive Learning

- Pull similar data points closer and push dissimilar ones further apart

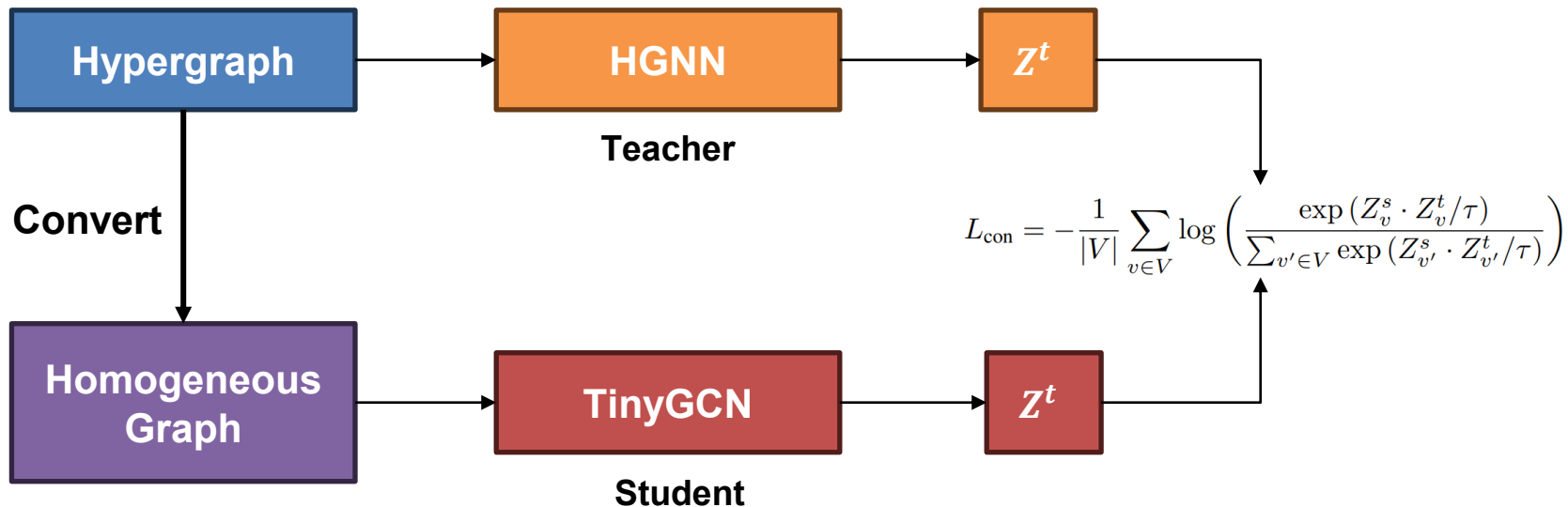


- Distill both embeddings and soft label results



Teacher – Contrastive Learning

- Pull vertices closer to each other in the embedding space



Teacher – Loss Function



$$L_{\text{bpr}} = \frac{1}{|V^L|} \sum_{v \in V^L} (Y_v - Y_v^t)^2$$

Difference between the true label and the soft labels

$$L_{\text{teacher}} = L_{\text{bpr}} + \gamma L_{\text{con}}$$

Add contrastive learning loss and label difference

Student – Loss Function



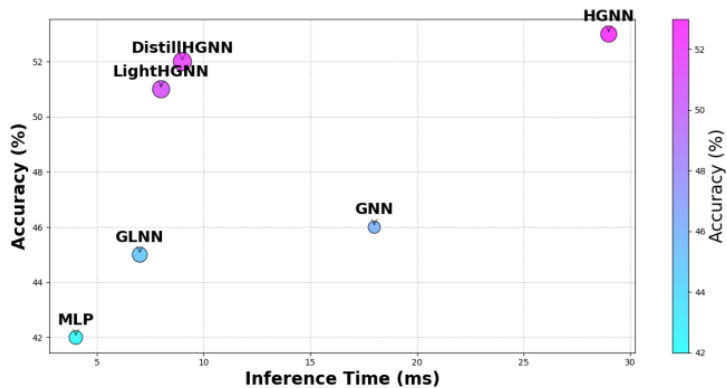
$$L_{\text{student}} = \frac{1}{|V^L|} \sum_{v \in V^L} (\hat{Y}_v - Y_v)^2 + \lambda \frac{1}{|V|} \sum_{v \in V} \text{KL}(\hat{Y}_v || Y_v^t)$$

Add true label difference loss and distillation loss

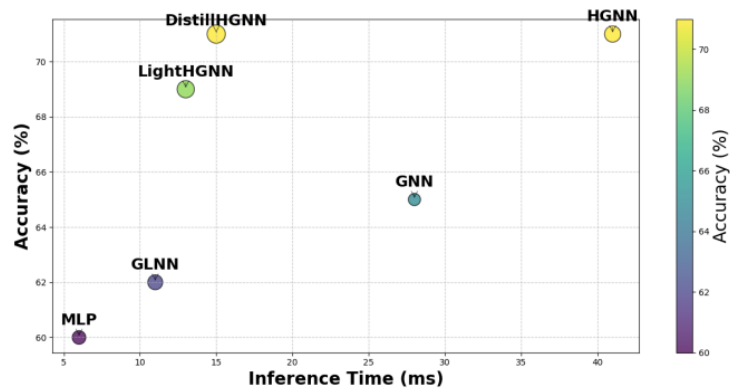
Experiments – Performance

Dataset	Setting	GNN	GCN	MLP	HGNN	GLNN	KRD	LightHGNN	DistillHGNN
IMDB	Tran.	46.45 ^{+2.15}	46.92 ^{+1.65}	43.22 ^{+1.95}	51.45 ^{+1.77}	45.88 ^{+2.33}	47.33 ^{+1.88}	50.22 ^{+1.95}	51.88^{+1.66}
	Ind.	47.33 ^{+2.44}	47.67 ^{+1.72}	44.15 ^{+2.12}	52.28 ^{+1.95}	46.85 ^{+2.55}	48.22 ^{+2.05}	51.18 ^{+2.12}	52.33^{+1.77}
	Prod.	46.88 ^{+2.33}	47.15 ^{+1.88}	43.77 ^{+2.05}	51.22 ^{+1.88}	46.12 ^{+2.44}	47.88 ^{+1.95}	50.45 ^{+2.05}	51.92^{+1.66}
IMDB-AW	Tran.	46.26 ^{+1.36}	46.77 ^{+2.12}	42.70 ^{+1.67}	53.36^{+2.25}	45.22 ^{+2.51}	47.15 ^{+1.88}	51.11 ^{+0.77}	52.34 ^{+1.62}
	Ind.	48.28 ^{+4.43}	48.67 ^{+2.33}	43.35 ^{+3.33}	54.12 ^{+2.74}	46.50 ^{+2.23}	49.12 ^{+2.55}	52.27 ^{+1.14}	55.27^{+1.16}
	Prod.	47.33 ^{+1.82}	47.88 ^{+1.95}	43.15 ^{+1.92}	53.31 ^{+3.01}	45.16 ^{+3.98}	48.22 ^{+2.15}	51.84 ^{+3.51}	53.93^{+2.14}
CC-CiteSeer	Tran.	53.06 ^{+0.62}	53.88 ^{+1.62}	46.76 ^{+0.83}	62.26^{+1.68}	52.57 ^{+1.73}	54.12 ^{+1.55}	59.65 ^{+2.12}	61.02 ^{+1.62}
	Ind.	55.00 ^{+1.09}	55.22 ^{+1.88}	49.20 ^{+2.60}	62.38^{+2.11}	54.45 ^{+3.01}	55.67 ^{+2.12}	61.36 ^{+1.27}	62.18 ^{+2.16}
	Prod.	53.60 ^{+0.34}	54.05 ^{+2.01}	47.20 ^{+1.60}	61.39 ^{+3.11}	52.08 ^{+2.55}	54.33 ^{+1.92}	60.11 ^{+1.63}	61.88^{+2.14}
CC-Cora	Tran.	52.26 ^{+3.25}	52.88 ^{+1.95}	45.16 ^{+2.51}	65.17^{+1.68}	51.22 ^{+1.10}	53.45 ^{+2.15}	63.65 ^{+2.12}	65.02 ^{+1.62}
	Ind.	54.39 ^{+2.15}	54.67 ^{+2.05}	48.66 ^{+2.19}	66.88^{+2.11}	53.50 ^{+1.81}	55.12 ^{+1.88}	65.76 ^{+1.27}	66.78 ^{+1.16}
	Prod.	54.15 ^{+3.44}	54.55 ^{+1.92}	48.02 ^{+2.05}	65.52 ^{+2.11}	53.19 ^{+2.75}	54.88 ^{+2.33}	64.11 ^{+1.63}	65.68^{+2.14}
Dataset	Setting	GNN	GCN	MLP	HGNN	GLNN	KRD	LightHGNN	DistillHGNN
DBLP	Tran.	75.33 ^{+2.15}	75.92 ^{+1.88}	66.88 ^{+2.77}	83.26^{+1.55}	72.45 ^{+2.55}	76.45 ^{+1.95}	81.45 ^{+2.88}	82.88 ^{+1.66}
	Ind.	76.15 ^{+2.44}	76.67 ^{+1.95}	67.22 ^{+3.15}	84.12 ^{+1.77}	73.12 ^{+2.88}	77.22 ^{+2.12}	82.33 ^{+2.55}	84.45^{+1.88}
	Prod.	75.88 ^{+2.33}	76.15 ^{+1.92}	66.45 ^{+2.95}	83.55 ^{+1.88}	72.88 ^{+2.66}	76.88 ^{+2.05}	81.88 ^{+2.44}	83.77^{+1.75}
DBLP-Paper	Prod.	65.15 ^{+2.21}	65.88 ^{+1.95}	60.47 ^{+2.44}	71.80^{+1.08}	62.42 ^{+4.02}	66.33 ^{+2.15}	69.12 ^{+3.22}	71.22 ^{+2.55}
	Tran.	66.23 ^{+1.55}	66.77 ^{+2.12}	62.38 ^{+2.45}	72.52 ^{+0.98}	63.87 ^{+3.76}	67.12 ^{+2.33}	71.22 ^{+2.63}	72.02^{+1.76}
	Ind.	65.57 ^{+1.86}	66.22 ^{+2.05}	61.50 ^{+3.07}	72.08 ^{+1.54}	63.17 ^{+3.22}	66.88 ^{+1.92}	70.69 ^{+2.17}	71.16^{+1.64}
DBLP-Term	Prod.	66.33 ^{+2.31}	66.92 ^{+1.88}	61.20 ^{+3.65}	72.10^{+0.90}	64.15 ^{+2.84}	67.45 ^{+2.15}	70.16 ^{+3.29}	71.67 ^{+0.87}
	Tran.	67.52 ^{+3.29}	67.88 ^{+2.15}	62.12 ^{+3.50}	73.46 ^{+1.63}	65.24 ^{+2.72}	68.33 ^{+2.44}	71.64 ^{+1.53}	73.79^{+1.32}
	Ind.	66.13 ^{+4.06}	66.77 ^{+2.05}	61.68 ^{+2.88}	73.12^{+1.33}	64.87 ^{+3.15}	67.22 ^{+2.33}	71.51 ^{+2.17}	72.45 ^{+1.76}
DBLP-Conf	Prod.	72.30 ^{+2.55}	72.88 ^{+1.95}	63.53 ^{+2.77}	90.03^{+1.66}	69.18 ^{+3.11}	73.45 ^{+2.15}	88.10 ^{+3.51}	89.27 ^{+2.17}
	Tran.	74.18 ^{+2.84}	74.67 ^{+2.12}	65.12 ^{+3.44}	92.22^{+2.78}	71.45 ^{+2.23}	75.22 ^{+2.33}	90.74 ^{+4.20}	92.11 ^{+1.84}
	Ind.	74.26 ^{+2.62}	74.88 ^{+2.05}	64.22 ^{+4.07}	91.00 ^{+2.42}	71.02 ^{+2.96}	75.33 ^{+1.92}	90.05 ^{+4.04}	91.38^{+3.25}

Experiments – Inference Time & Accuracy



(a) IMDB-AW



(b) DBLP

Experiments – High-Order Preservation

- The higher the High-Order Preservation Score is, the greater structural knowledge is preserved

Method	CC-Cora	CC-Citeseer	IMDB-AW	DBLP	Mean
GLNN (Tian et al., 2022)	0.54	0.58	0.51	0.67	0.575
KRD (Wu et al., 2023)	0.56	0.62	0.54	0.71	0.6075
LightHGNN (Feng et al., 2024)	0.72	0.78	0.74	0.83	0.7675
DistillHGNN	0.78	0.84	0.81	0.88	0.8275

$$H_{\text{pres}} = \frac{1}{|V|} \sum_{v \in V} \frac{1}{|N(v)|} \sum_{u \in N(v)} \text{Sim}(\mathbf{x}_u^{\text{teacher}}, \mathbf{x}_u^{\text{student}})$$

- Hypergraphs often outperform graphs on higher-order interactions, but are computationally expensive - distillation is needed for efficient inference.
- LightHGNN performs logits distillation from an HGNN teacher to an MLP student, with a high-order hyperedge preservation loss
- DistillHGNN aligns teacher–student embeddings via contrastive learning, then applies an MLP head and logits distillation to preserve high-order information